Basics

Tuesday, November 25, 2025 10:49 PM

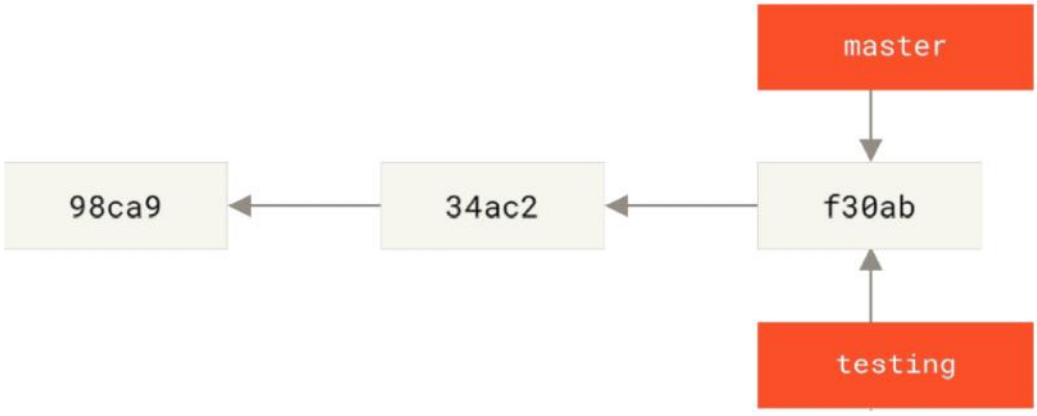
Creating a branch

git branch <branch_name>

git branch testing

To create a new branch and switching to it at the same time

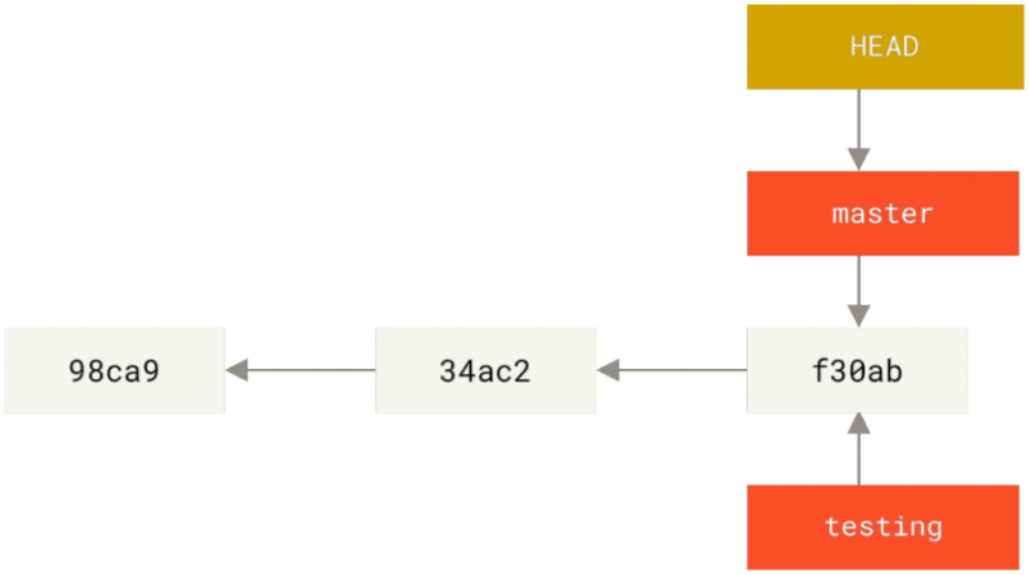
git checkout -b <branch_name>



How does Git know what branch you're currently on?

1. It keeps a special pointer called **HEAD**.
2. In Git, this is a **pointer** to the **local branch** you're currently on.
3. In this case, you're on master.

Note: The *git branch* command **only created** a new branch — it **didn't switch** to created branch



Let's see this by running a simple git log command that shows you where the branch pointers are pointing. This option is called –decorate

git log --online --decorate

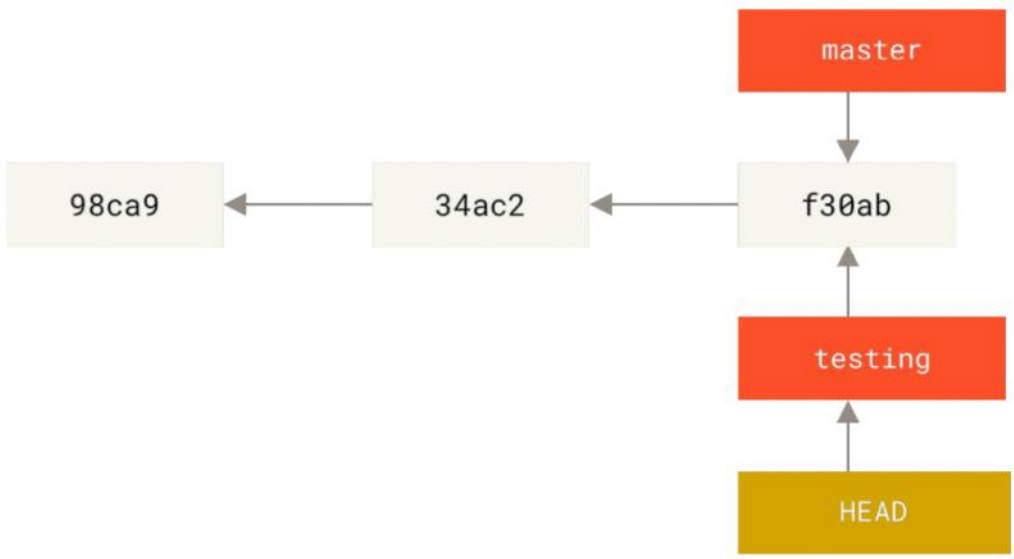
```
$ git log --oneline --decorate
09b870a (HEAD -> master, tag: show, testing) Merge branch 'main' of https://github.com/msajidaligik/temp
d64fda1 (origin/main, main) Create temp
4686da5 changes in gitignore
a4174a0 some changes in gitignore file
08aeb80 (tag: v1.0.1) initial commit
```

Switching a branch

git checkout <branch_name>

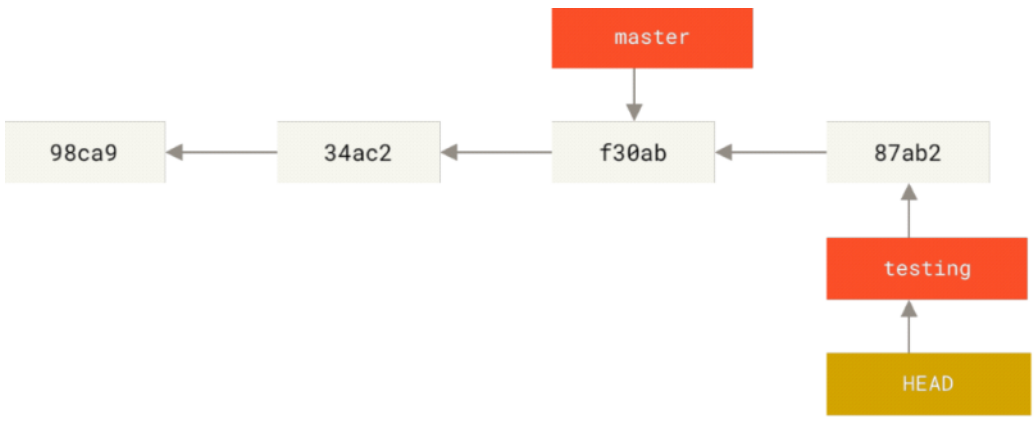
git checkout testing

Note: This will **move** the **HEAD** pointer to the **testing branch**



<u>Well, let's do some changes in new switched branch</u>

1. Modify file
2. Stage the modified file
3. Commit the changes




```
$ git log --oneline --decorate --graph --all
* 6f364f7 (HEAD -> testing) some more changes in README.MD file
*   09b870a (tag: show, master) Merge branch 'main' of https://github.com/msajidaligik/temp
|\
| * d64fda1 (origin/main, main) Create temp
* 4686da5 changes in gitignore
* a4174a0 some changes in gitignore file
* 08aeb80 (tag: v1.0.1) initial commit
```

This is **interesting**, because now your **testing** branch has **moved forward**, but your **master** branch **still points** to the **commit** you were on when you **ran git checkout** to switch branches.

Let's switch back to the master branch:

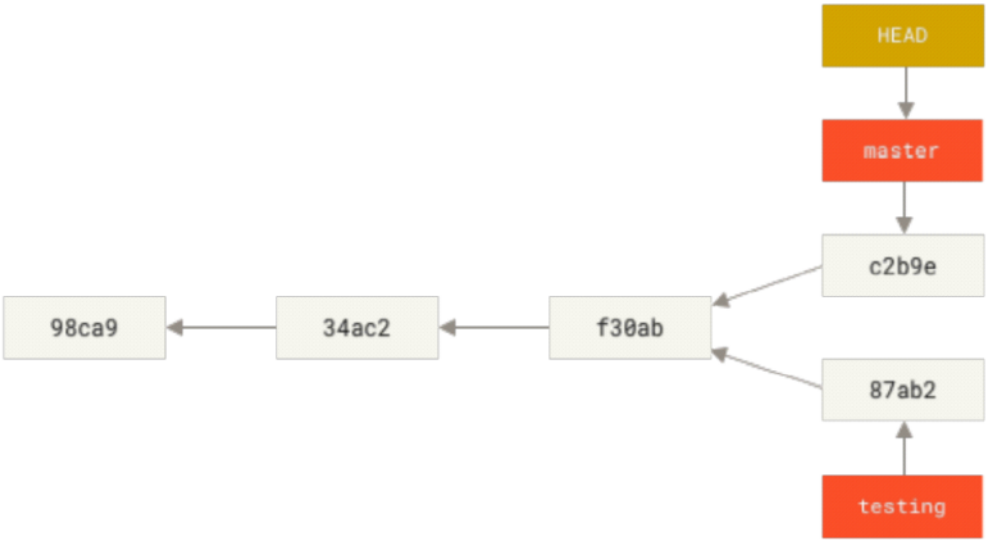
git checkout master

Note: this command did two things

1. **Moved HEAD** pointer to the master branch
2. R**everted files** in the working directory to the snapshot of master branch

Do Some More Changes in Master Branch:

1. Create a new file
2. Stage it
3. Commit the changes of working directory




```
$ git log --oneline --decorate --graph --all
* 3ebb65a (HEAD -> master) new file added
| * 6f364f7 (testing) some more changes in README.MD file
|/
*   09b870a (tag: show) Merge branch 'main' of https://github.com/msajidaligik/temp
|\
| * d64fda1 (origin/main, main) Create temp
* 4686da5 changes in gitignore
* a4174a0 some changes in gitignore file
* 08aeb80 (tag: v1.0.1) initial commit
```

This is called **divergent history**